

DebtStream

Digital Collections Analysis

Digital Collections Funnel and Payment Journey Optimisation
End-to-End Data Analytics Project Report

BigQuery | Python | Power BI

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1. Executive Summary

This report presents an end-to-end analytics project for DebtStream, a B2B SaaS fintech platform that moves customer repayment journeys into a digital self-service flow. The project analyses 10,000 customer accounts uploaded between June 2022 and May 2023 to understand how customers move through the digital collections journey, how SMS message tone affects repayment behaviour, and which account segments create the greatest recovery risk.

The analytical pipeline uses BigQuery SQL for raw data validation, customer-level analytical view construction and summary output generation, Python for exploratory data analysis, statistical significance testing and recovery insight generation, and Power BI for a compact three-page interactive stakeholder dashboard.

Headline Performance Metrics

Metric	Value	Context
Total Customer Accounts	10,000	Accounts uploaded June 2022 to May 2023
Total Outstanding Debt Balance	£9,996,602	Total debt balance across all customer accounts
Total Amount Collected	£330,918	Successful payments only; failed and reversed excluded
Overall Recovery Rate	3.31%	Amount collected divided by outstanding balance
Invite Delivered Rate	97.72%	9,772 of 10,000 customers received their invite
Digital Journey Start Rate	37.81%	3,695 of 9,772 invited customers clicked the link
Payment Conversion Rate	11.36%	1,136 of 10,000 customers made a successful payment
Repayment Plan Setup Rate	7.43%	743 of 10,000 customers created a repayment plan
Broken Plan Rate	40.51%	301 of 743 repayment plans subsequently broken
A/B Test: Variant Plan Setup Lift	+4.39pp	Supportive SMS outperformed direct reminder on plan setup
A/B Test: Variant Amount Lift	+£6.40	Variant customers collected £6.40 more per customer on average
Affordability Completion Rate	2.03%	203 of 10,000 customers completed the affordability assessment

The audit identifies three primary strategic findings. First, the largest drop-off in the digital journey occurs at the Invite-to-Link-Opened transition, where 62.19% of customers who received an invite never clicked through. This is the single biggest lever for improving collection outcomes. Second, the SMS A/B test confirms that supportive repayment messaging materially outperforms a direct payment reminder across every key metric except one-off payment rate, with Variant customers generating £6.40 more per customer on average and setting up repayment plans at nearly double the rate. Third, affordability completion has a measurable positive impact on plan quality: customers who completed affordability assessment paid their first instalment at 86.21% vs 75.19%, and broke their plans at 31.53% vs 43.89%.

Primary Recommendation: DebtStream should adopt a segmented collections strategy. The SMS A/B test supports prioritising supportive repayment-options messaging for medium and higher-balance customers who need flexibility, while maintaining direct payment reminders for customers with higher immediate payment readiness. Reducing the Invite-to-Link-Opened drop-off and scaling affordability assessment completion are the two highest-impact operational interventions identified in this dataset.

2. Project Background and Objectives

Business Context

DebtStream is a B2B SaaS and fintech collections platform that helps lenders, utility providers, telecoms companies, debt purchasers, debt collection agencies and legal collections firms move customer repayment journeys into a digital self-service model. The business value of the platform depends on more than sending reminders. It depends on whether customers open the journey, verify their identity, view their outstanding balance, select a repayment option, make a successful payment, and sustain a repayment plan over time.

Business Questions

- Where do customers drop off in the digital collections funnel, and which stage creates the biggest customer loss?
- Which primary communication channels drive customers into the self-service journey most effectively?
- Did the supportive SMS repayment messaging strategy (Variant) outperform the direct payment reminder (Control) in the A/B test, and for which balance segments?
- Which balance bands and debt age bands recover better or worse through the digital collections process?
- Does completing the affordability assessment improve repayment plan sustainability and reduce broken-plan rates?
- Which customer segments combine weak recovery rates with high outstanding balance exposure?
- What should DebtStream prioritise to improve payment conversion, plan sustainability, and portfolio recovery outcomes?

Project Scope and Pipeline

Phase	Tool	Key Deliverable
Phase 1	BigQuery SQL	Raw table validation, relationship checks, funnel analysis, SMS A/B test outputs, recovery analysis and Power BI reporting views
Phase 2	Python (Pandas, Matplotlib, Seaborn)	Data validation, EDA, statistical significance testing, recovery insights, exported charts and summary tables
Phase 3	Power BI Desktop	Three-page interactive dashboard covering funnel performance, SMS A/B test outcomes and recovery risk segments

3. Data Overview

The dataset contains six core relational tables ingested into BigQuery, covering B2B client organisations, customer accounts, outbound communications, digital journey events, payment transactions, and repayment plans. The project period covers account uploads from June 2022 to May 2023. Downstream journey and payment events may extend beyond the upload month because customers can engage and pay after the initial communication date.

Core Raw Tables

Table	Records	Key Columns	Description
clients	10	client_id, client_type, industry, integration_type, client_size	B2B client metadata including client type, industry and integration details
customers	10,000	customer_id, client_id, debt_balance, balance_band, debt_age_band, portfolio_type, account_upload_date	Customer account-level data including upload date, balance band, debt age, region and portfolio type
communication_events	19,244	customer_id, channel, sent_date, delivered_flag, opened_or_clicked_flag, ab_test_group	SMS, email, letter and agent referral events with delivery, click/open and A/B test group fields
journey_events	50,699	customer_id, event_name, event_date	Digital self-service events including link open, identity verification, balance view, payment option selection, and plan creation
payments	2,362	customer_id, payment_date, payment_amount, payment_status, payment_type	Payment records with status, amount, method and timing
repayment_plans	743	customer_id, plan_created_date, plan_status, first_instalment_paid_flag, broken_plan_flag, affordability_completed_flag	Plan records including first instalment status, missed payments, affordability flag and plan outcome

BigQuery Analytical Views

Seven views were built on top of the raw tables to serve as the primary analytical inputs for Python and Power BI. Only three views are imported into Power BI as the reporting layer; the remaining four are used for Python EDA only.

View	Grain	Purpose	Used In
vw_customer_summary	1 row per customer	Master fact table: all funnel flags, segments, payment, plan, affordability and recovery fields joined from all 6 raw tables	Power BI Fact Customer, Python EDA
vw_ab_test_customer_outcomes	1 row per A/B customer	Filtered to 1,936 SMS A/B test customers with experiment group, behavioural flags and commercial outcomes	Power BI Fact AB Test, Python A/B analysis
vw_funnel_summary	1 row per month x segment	Aggregated funnel rates by upload month, channel, balance band, debt age and portfolio type	Power BI Monthly Funnel Trend
vw_channel_performance	Aggregated	Channel-level funnel and recovery summary	Python EDA only
vw_ab_test_summary	Aggregated	Group-level A/B test outcome summary	Python EDA only
vw_recovery_summary	Aggregated	Recovery rates by balance band, debt age and portfolio type	Python EDA only
vw_plan_performance	Plan level	Plan sustainability with affordability_status CASE label	Python EDA only

Key Analytical Dimensions

Dimension	Values
Analysis Period	June 2022 to May 2023 (12 months)
Balance Bands	Low (3,508), Medium (4,065), High (1,853), Very High (574)
Debt Age Bands	0-30 (1,539), 31-90 (2,123), 91-180 (2,842), 181-365 (2,224), 365+ (1,272)
Primary Channels	SMS (5,922 customers, 59.22%), Email (3,015, 30.15%), Letter (1,058, 10.58%), Agent Referral (5, less than 0.1%)
Client Types	Utility Provider, Consumer Lender, Debt Collection Agency, Debt Purchaser, Telecoms, Legal Collections
Portfolio Types	Early Arrears, Late Arrears, Recoveries, Purchased Debt, Legal Collections
A/B Test Groups	Control (929 customers, direct payment reminder), Variant (1,007 customers, supportive repayment-options message)
Affordability Status	Affordability Completed (203 customers), No Affordability Completed (9,797 customers)

4. Methodology

4.1 Funnel Analysis

The digital collections funnel was calculated using distinct customer_id counts at each stage to avoid inflating results from repeated communications or repeated journey events. Nine stages were measured: Account Uploaded, Invite Sent, Invite Delivered, Link Opened, Identity Verified, Balance Viewed, Payment Option Selected, Payment Made, and Repayment Plan Created. Stage-to-stage drop-off rates were calculated as the percentage of customers at the prior stage who did not progress to the next. Channel-level journey start rates were calculated by filtering the customer base by primary communication channel.

4.2 SMS A/B Test Analysis

The A/B test compared a direct payment reminder (Control, 929 customers) against a supportive repayment-options message (Variant, 1,007 customers). The unit of analysis was the customer, not the communication event, to avoid double-counting customers who received multiple reminders. Five binary outcomes were compared: Link Open Rate, Payment Rate, One-off Payment Rate, Plan Setup Rate, and First Instalment Paid Rate. Python was used to run two-proportion z-tests for all binary outcomes and a Welch t-test for amount collected per customer to confirm statistical significance of observed differences. Lift was calculated as Variant rate minus Control rate.

4.3 Recovery and Segment Analysis

Recovery rate was calculated at individual customer level as successful amount collected divided by total outstanding balance. Failed and reversed payments were excluded from amount collected. Recovery outcomes were segmented by balance band, debt age band, client type, portfolio type, and affordability status. A balance band by debt age band recovery matrix was produced to identify the weakest-performing segment combinations and highest-priority intervention areas.

4.4 Power BI Data Model

Three BigQuery views were imported into Power BI Desktop: vw_customer_summary as Fact Customer (10,000 rows), vw_ab_test_customer_outcomes as Fact AB Test (1,936 rows), and vw_funnel_summary as Monthly Funnel Trend (4,360 rows). Three disconnected helper tables were created using DAX DATATABLE: Funnel Stages (8 rows), AB Core Metrics (5 rows), and AB Lift Metrics (5 rows). These helper tables drive the dynamic funnel and A/B test charts via SWITCH measures. Fact Customer and Fact AB Test are deliberately kept disconnected because the A/B test is a filtered experiment population. 60 DAX measures were built and validated against known benchmark values from SQL outputs.

Method	Why It Was Used
Distinct customer-level funnelling	Prevents repeated communications or journey events from inflating funnel counts

Method	Why It Was Used
Stage drop-off analysis	Identifies which journey steps create the largest behavioural friction and customer loss
Two-proportion z-tests	Tests whether Control vs Variant outcome differences are statistically meaningful, not just directional
Recovery segmentation	Connects repayment outcomes to balance band, debt age, client type and affordability status
Disconnected DAX SWITCH tables	Enables dynamic funnel and A/B visuals in Power BI without requiring relationships between experiment and customer tables

5. Phase 1: BigQuery SQL: Data Validation and Analytical Views

The SQL phase established the analytical foundation in BigQuery across five files executed in sequence. Each file answered specific business questions and produced validated outputs for downstream Python analysis and Power BI reporting.

5.1 SQL File Structure

SQL File	Purpose
01_Setup_and_Base_Tables.sql	Validates ingestion of all 6 raw tables: row counts, date ranges, duplicate ID checks, referential integrity, numeric sanity checks and category distributions
02_Funnel_and_Channel_Analysis.sql	Builds the customer-level funnel, stage-to-stage drop-off rates, and funnel performance segmented by channel, balance band, debt age band, client type and upload month
03_SMS_AB_Testing.sql	Creates vw_ab_test_customer_outcomes, compares Control vs Variant across 5 outcome metrics, segments A/B performance by balance band and client type, and produces lift summary
04_Repayment_and_Recovery.sql	Analyses payment conversion, amount collected, recovery rate segmented by balance band and debt age, plan sustainability, affordability assessment impact and priority recovery segments
05_PowerBI_Views.sql	Creates all 7 BI-ready views for Power BI including vw_customer_summary, vw_funnel_summary, vw_channel_performance, vw_ab_test_summary, vw_recovery_summary, vw_client_portfolio_summary and vw_plan_performance

5.2 Database Schema

A relational schema was designed in BigQuery with six core tables. Each table was validated for row counts, duplicate IDs, referential integrity across all foreign key relationships, numeric field sanity, and categorical value distributions before any analytical work began.

Table	Primary Key	Relationships	Purpose
clients	client_id	None	Client organisation attributes, industry, integration type, client size
customers	customer_id	client_id to clients	Customer demographics, debt balance, segments, upload date, portfolio type

Table	Primary Key	Relationships	Purpose
communication_events	communication_id	customer_id to customers	Invite and SMS send, delivery, open/click events, A/B test group assignment
journey_events	event_id	customer_id to customers	Digital self-service journey event records
payments	payment_id	customer_id to customers	Payment transactions, amounts, type flags, payment status
repayment_plans	plan_id	customer_id to customers	Plan creation, instalment, affordability completion and broken-plan flags

5.3 Key SQL Exports

The following CSV files were exported from BigQuery for use in Python analysis and Power BI validation:

- overall_funnel.csv and stage_drop_off.csv for the end-to-end funnel and drop-off rates
- funnel_by_channel.csv, funnel_by_balance_band.csv, funnel_by_debt_age.csv and funnel_by_client_type.csv for funnel segmentation
- ab_test_customer_outcomes.csv, ab_test_core_performance.csv, ab_test_lift_summary.csv and ab_test_by_balance_band.csv for SMS A/B test analysis
- customer_recovery_summary.csv, recovery_by_balance_band.csv, recovery_by_debt_age.csv, recovery_by_client_type.csv and recovery_priority_segments.csv for recovery analysis

6. Phase 2: Python: EDA, A/B Testing and Recovery Analysis

All BigQuery views and CSV summary exports were loaded into Pandas DataFrames for exploratory data analysis, statistical testing and business insight generation. Three analytical notebooks were produced in sequence, each building on the outputs of the previous phase.

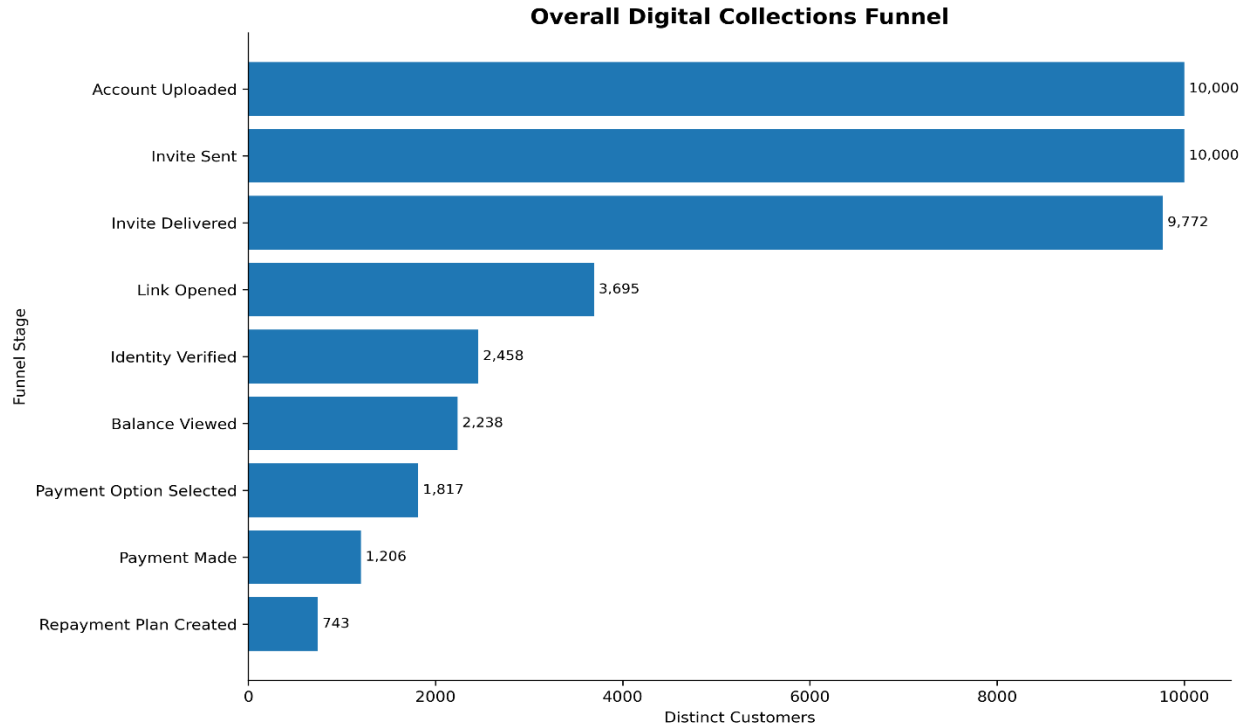
Notebook	Purpose
01_Data_Validation_and_EDA.ipynb	Validates raw data structure, explores missing values and date ranges, analyses customer distribution by segment, visualises the overall digital collections funnel and monthly trends
02_SMS_AB_Testing_Analysis.ipynb	Compares Control vs Variant SMS performance across 5 outcome metrics, runs two-proportion z-tests for statistical significance and recommends a segmented messaging strategy
03_Repayment_and_Recovery_Insights.ipynb	Analyses recovery baseline, recovery performance by balance band, debt age band and client type, affordability assessment impact on plan sustainability, and priority recovery segments

6.1 Notebook 1: Data Validation and EDA

Notebook 1 confirmed that all six raw datasets loaded successfully and matched the BigQuery-validated row counts: 10,000 customers, 19,244 communication events, 50,699 journey events, 2,362 payments and 743 repayment plans. Data quality was strong overall with no critical cleaning required.

Chart 1: Overall Digital Collections Funnel

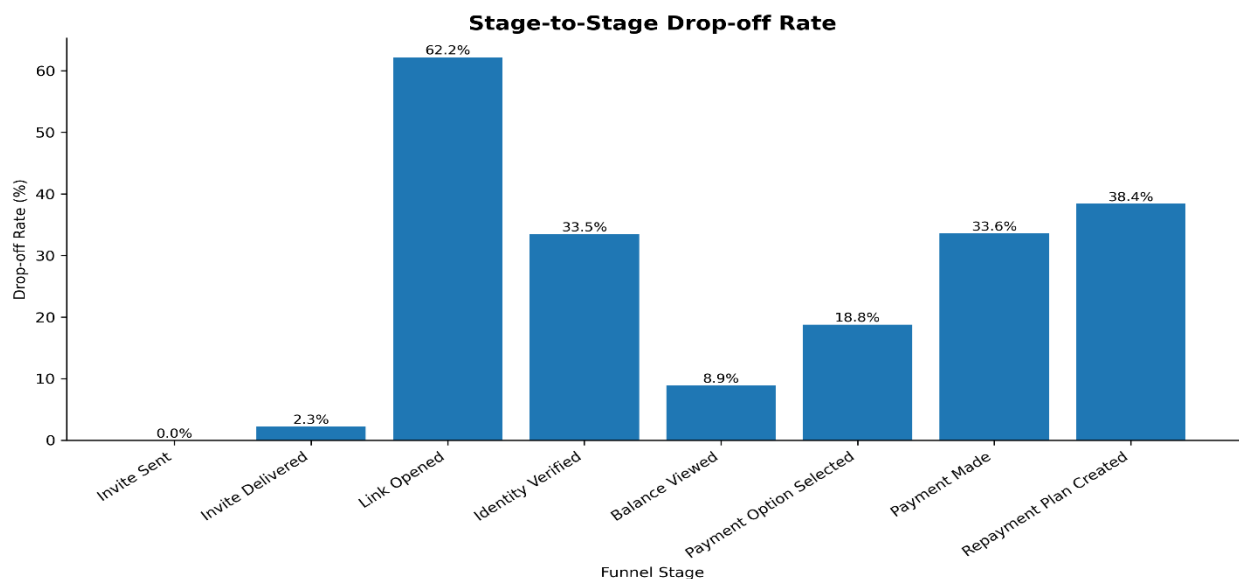
The nine-stage digital journey was visualised showing absolute customer counts at each stage. The Invite Delivered to Link Opened transition was confirmed as the single largest drop-off, with 6,077 customers (62.19% of those who received an invite) never clicking through to the digital journey.



Business Takeaway: The Invite-to-Link-Opened drop-off of 62.19% is the largest single conversion failure in the entire customer journey. Improving message engagement at this stage creates a larger opportunity than optimising any later step in the journey.

Chart 2: Stage-to-Stage Drop-off Rate

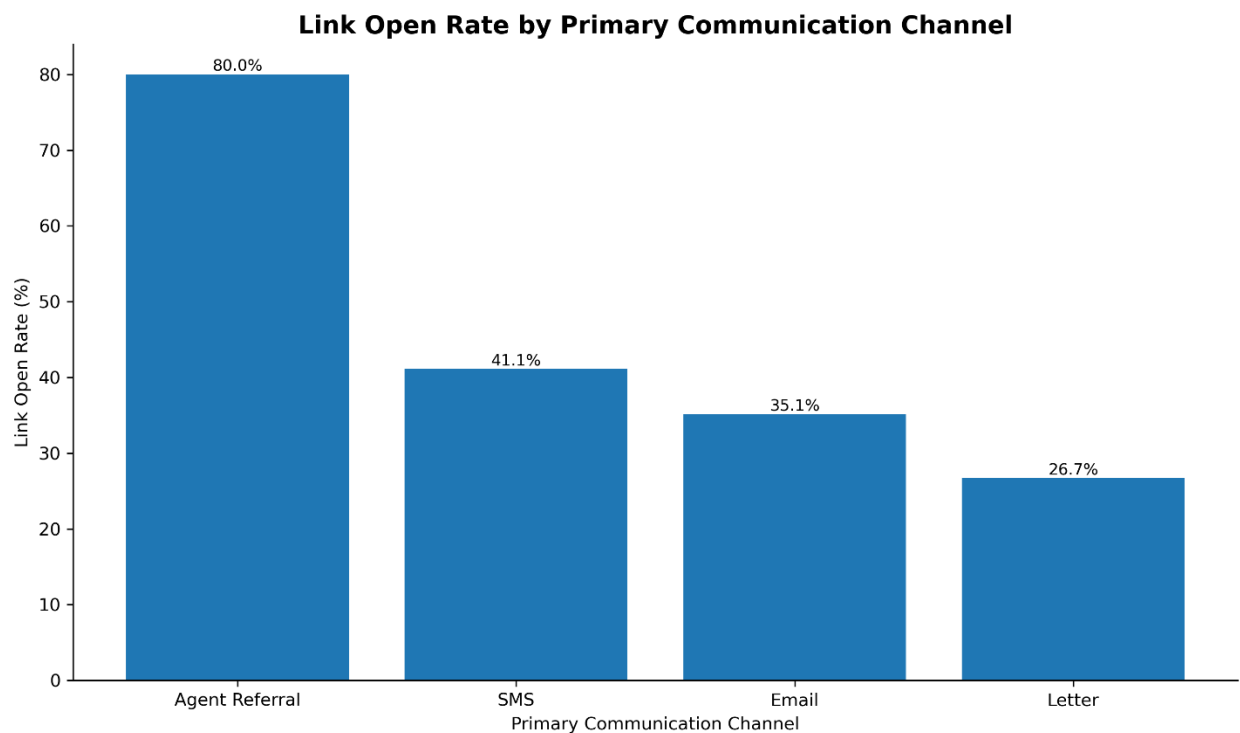
Stage-to-stage drop-off percentages were plotted to identify where friction is highest. The chart confirms that after Link Opened, the journey retains customers at a far higher rate, reinforcing that the primary challenge is journey entry rather than journey completion.



Business Takeaway: Journey optimisation should focus first on improving message engagement to drive link opens, then on reducing friction at identity verification which is the second-largest drop-off point.

Chart 3: Digital Journey Start Rate by Primary Channel

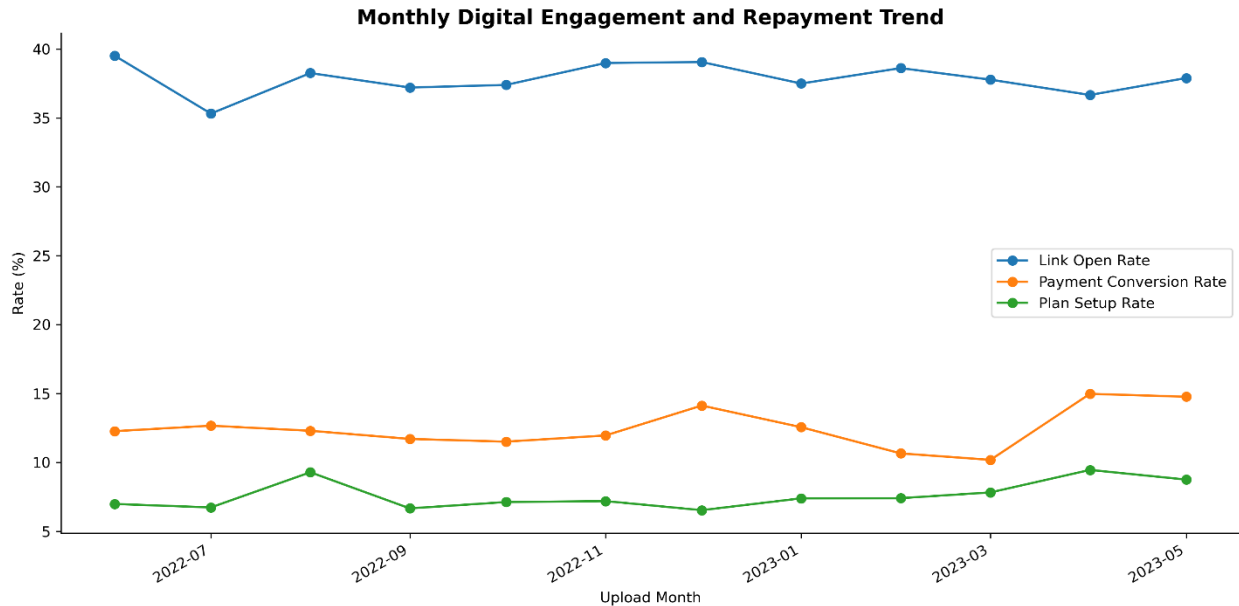
Journey start rate (link opened divided by invite delivered) was compared across channels: SMS (41.11%), Email (35.14%), and Letter (26.70%). Agent Referral produced a high rate at 80% but with only 5 customers, making it statistically insignificant as a scalable channel.



Business Takeaway: SMS-first communication strategies produce the highest digital engagement rates at scale. Letter-based communication achieves the lowest start rate, suggesting that post-letter SMS follow-up could meaningfully improve funnel entry for letter-contacted customers.

Chart 4: Monthly Funnel Trend

Link open rate, payment conversion rate, and plan setup rate were plotted month by month across the full June 2022 to May 2023 period. Link open rates remained stable at approximately 36 to 39%, with payment conversion rates ranging from 9.3% to 13.8% and plan setup rates from 6.4% to 9.3%.

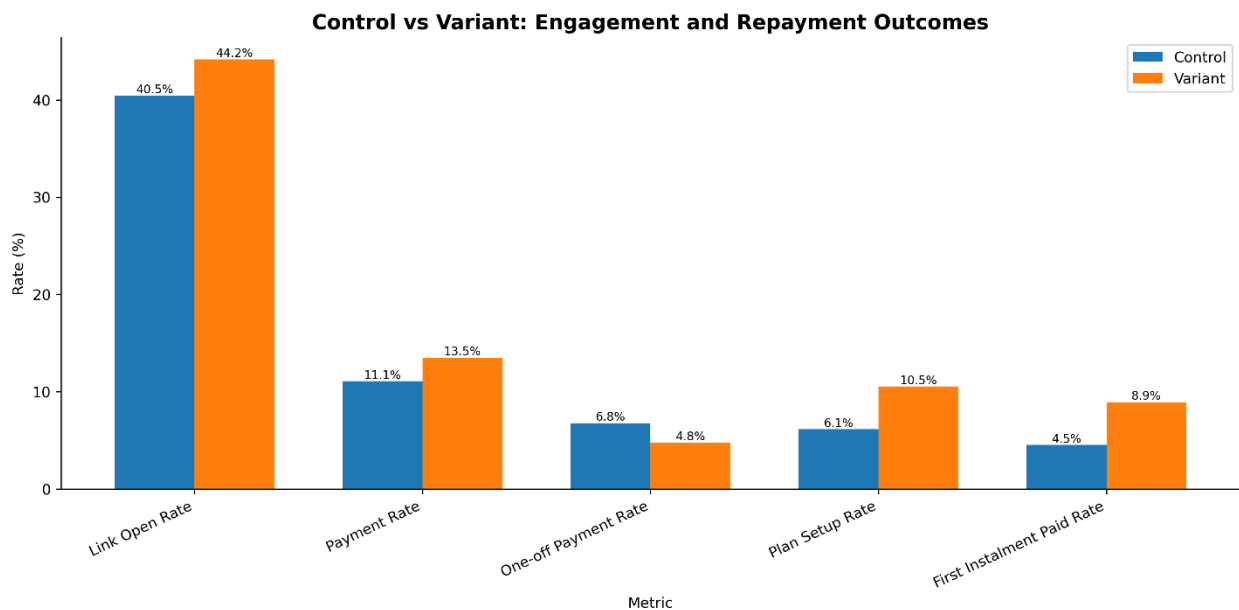


Business Takeaway: April and May 2023 show a notable uptick in both payment conversion and plan setup rates. This suggests possible seasonal effects or operational improvements in the final two months of the analysis window that warrant closer investigation.

6.2 Notebook 2: SMS A/B Testing Analysis

Chart 5: Control vs Variant Core Outcome Rates

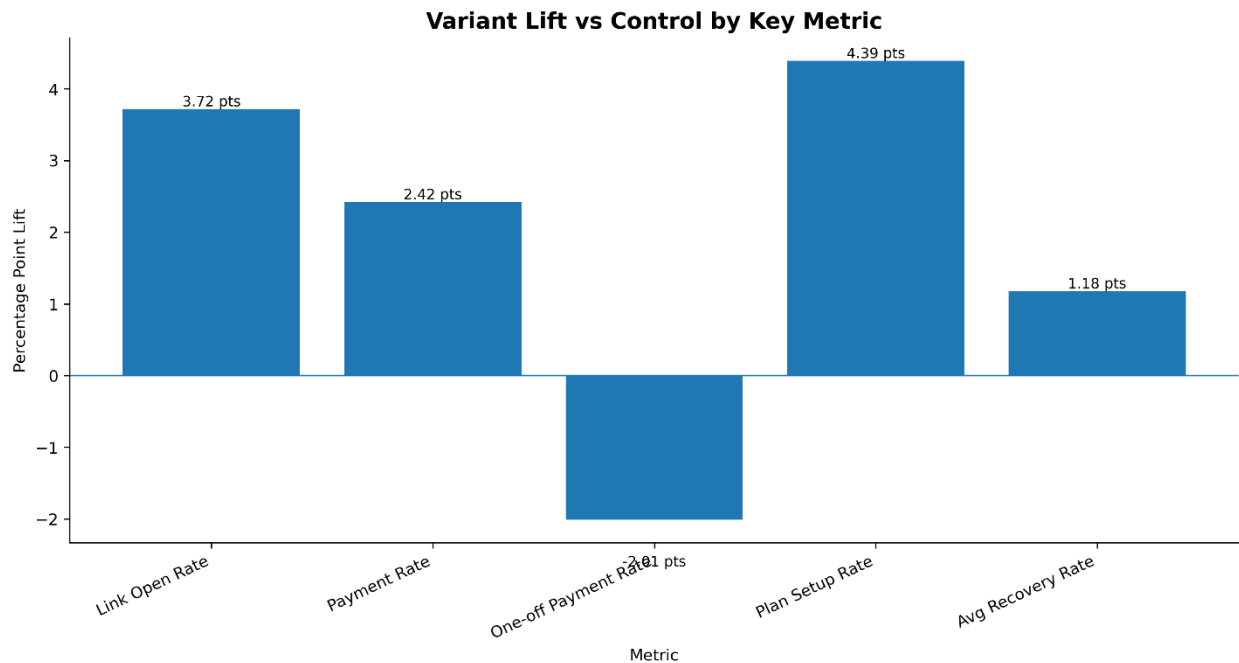
Five key outcome metrics were compared side by side between Control and Variant groups. Variant outperformed Control on Link Open Rate (+3.72pp), Payment Rate (+2.42pp), Plan Setup Rate (+4.39pp), and First Instalment Paid Rate (+11.22pp). Control outperformed Variant on One-off Payment Rate (+2.01pp).



Business Takeaway: Supportive messaging shifts customers toward more sustainable plan-based repayment rather than simply changing how they pay. The Variant group's superior plan setup rate (+4.39pp) and first instalment paid rate (+11.22pp) confirm that the approach produces higher-quality, more durable collection outcomes.

Chart 6: Variant Lift vs Control by Metric

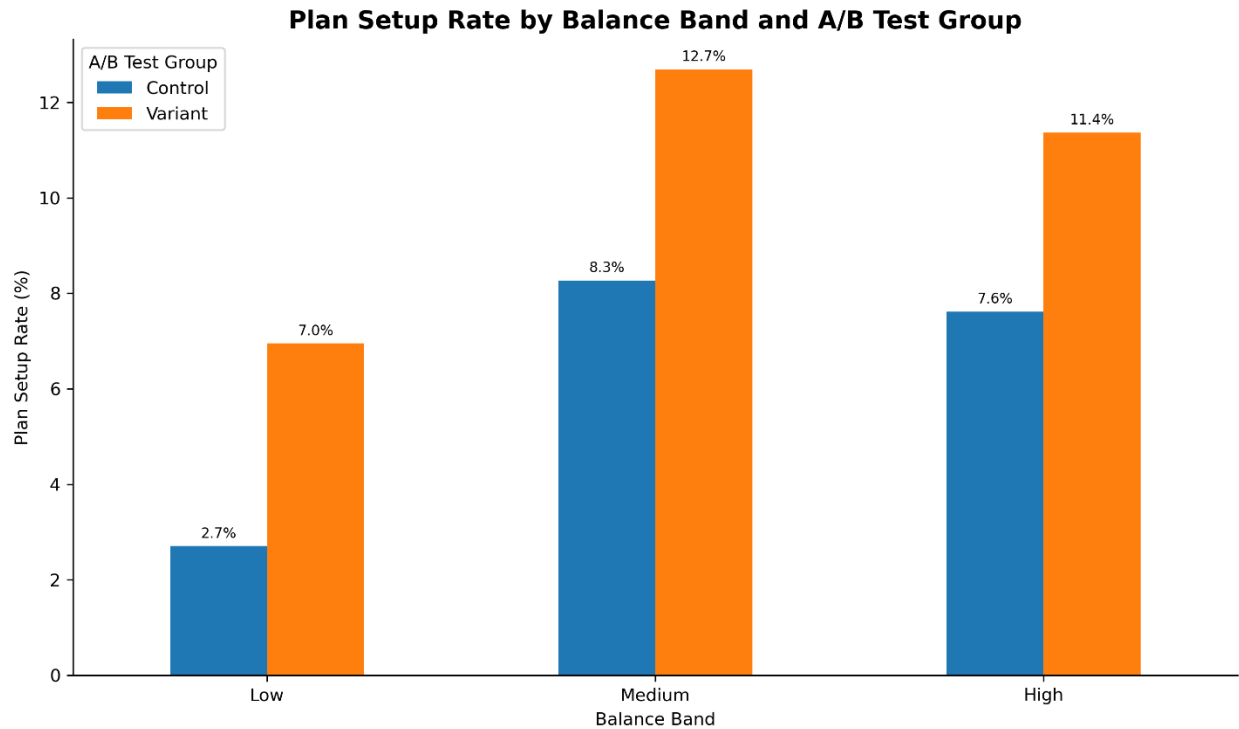
Lift values (Variant minus Control) were visualised to show positive and negative outcomes clearly. Plan Setup Rate Lift (+4.39pp) was the strongest positive outcome. One-off Payment Rate Lift (-2.01pp) was the only negative, confirming that the supportive message reduces one-off payments in favour of structured plan setups.



Business Takeaway: The negative one-off payment lift is not a failure of the Variant approach. It reflects a deliberate shift in customer behaviour toward structured repayment plans, which is a commercially superior outcome for long-term portfolio recovery. Statistical significance testing confirmed that plan setup lift and first instalment paid rate lift are statistically significant (p less than 0.05) overall, and especially for Low and Medium balance segments.

Chart 7: Plan Setup Rate by Balance Band, Control vs Variant

Variant outperforms Control on plan setup rate across all three tested balance bands. The gap is widest in the Medium balance segment (Control 8.27% vs Variant 12.69%, a 4.42pp lift) and narrowest in the High balance segment (Control 7.62% vs Variant 11.37%, a 3.75pp lift). Very High balance customers are not represented in the A/B test population.

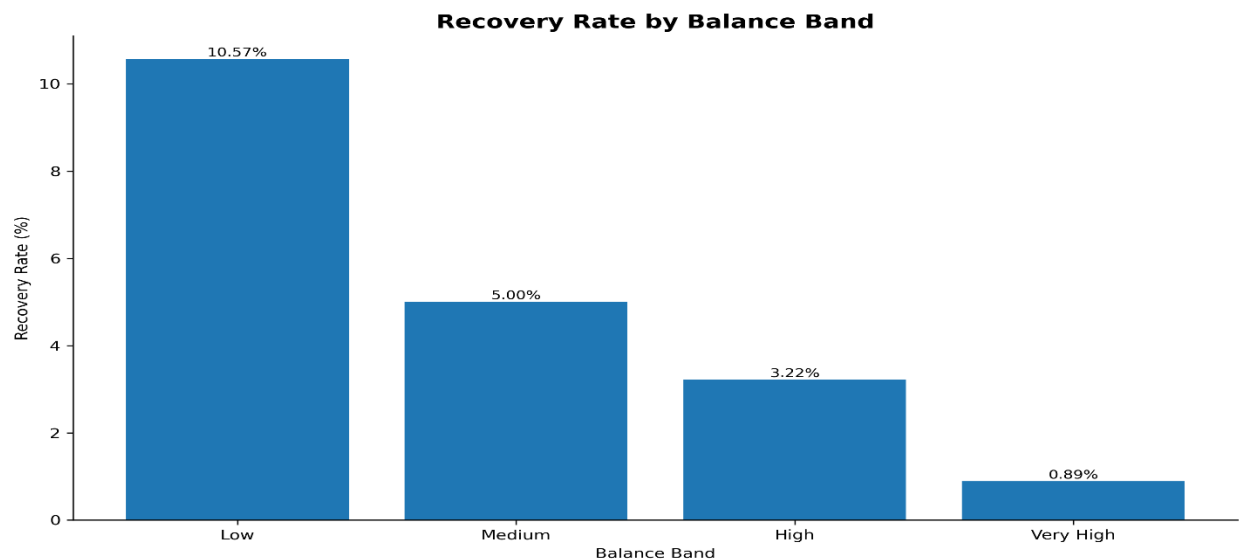


Business Takeaway: The supportive messaging strategy works consistently across all balance segments tested. Medium balance customers show the strongest absolute response and should be prioritised for immediate rollout of the Variant approach.

6.3 Notebook 3: Repayment and Recovery Insights

Chart 8: Recovery Rate by Balance Band

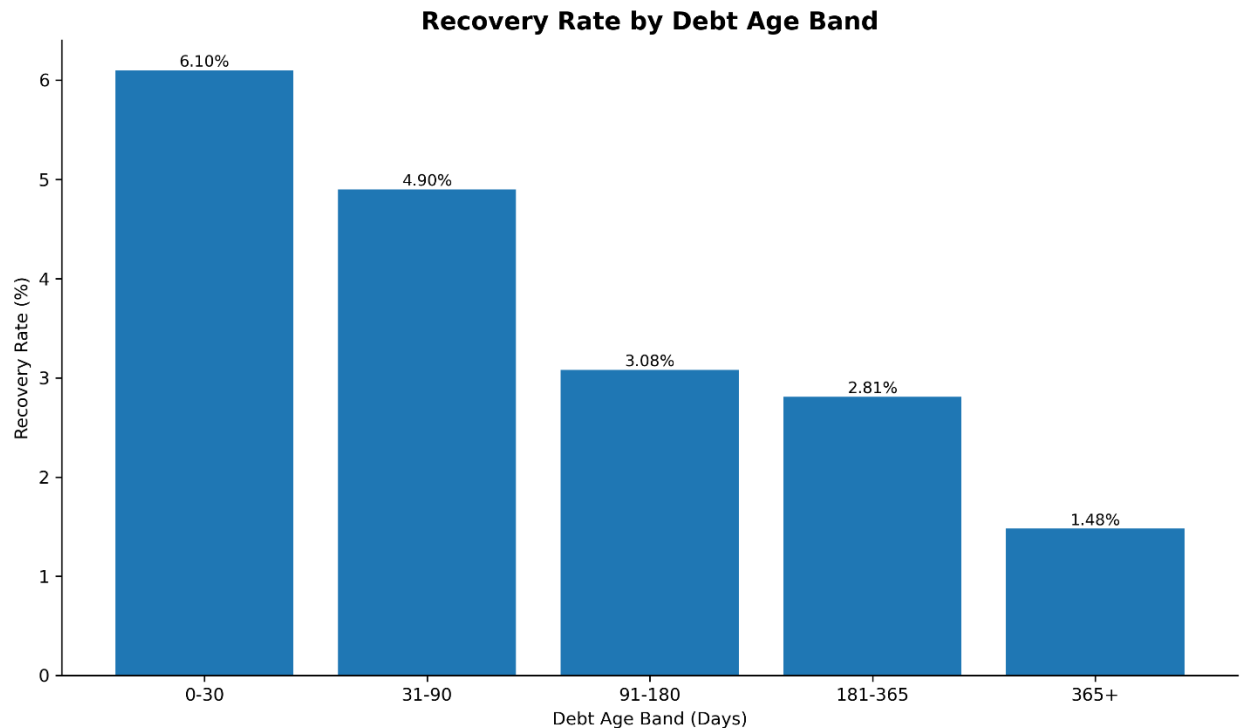
Low balance customers recover at the highest rate (10.57%), declining sharply through Medium (5.00%), High (3.22%), and Very High (0.89%). Very High balance accounts have an outstanding balance of £3,192,271 but a broken-plan rate of 72.22%, representing the highest combined commercial risk and plan sustainability problem in the portfolio.



Business Takeaway: DebtStream should treat balance band as a core segmentation variable. Low balances suit direct payment journeys, Medium balances suit repayment-plan journeys, and High or Very High balances need affordability-led support and closer plan monitoring because recovery is weaker and broken-plan risk is substantially higher.

Chart 9: Recovery Rate by Debt Age Band

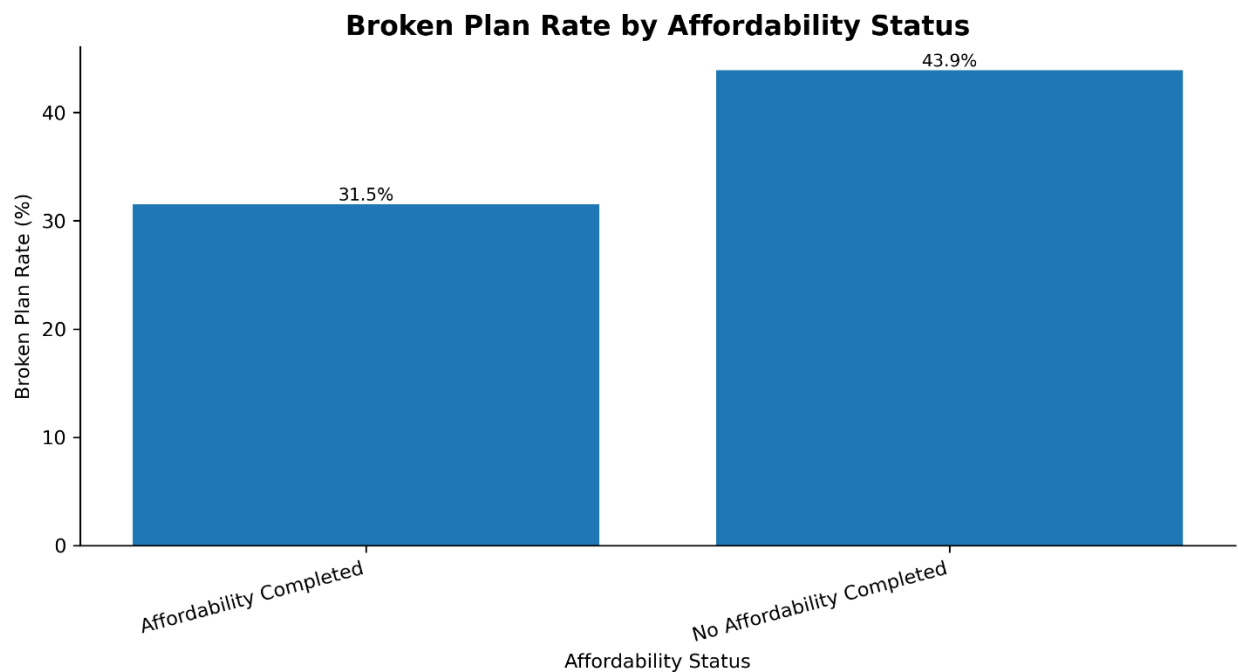
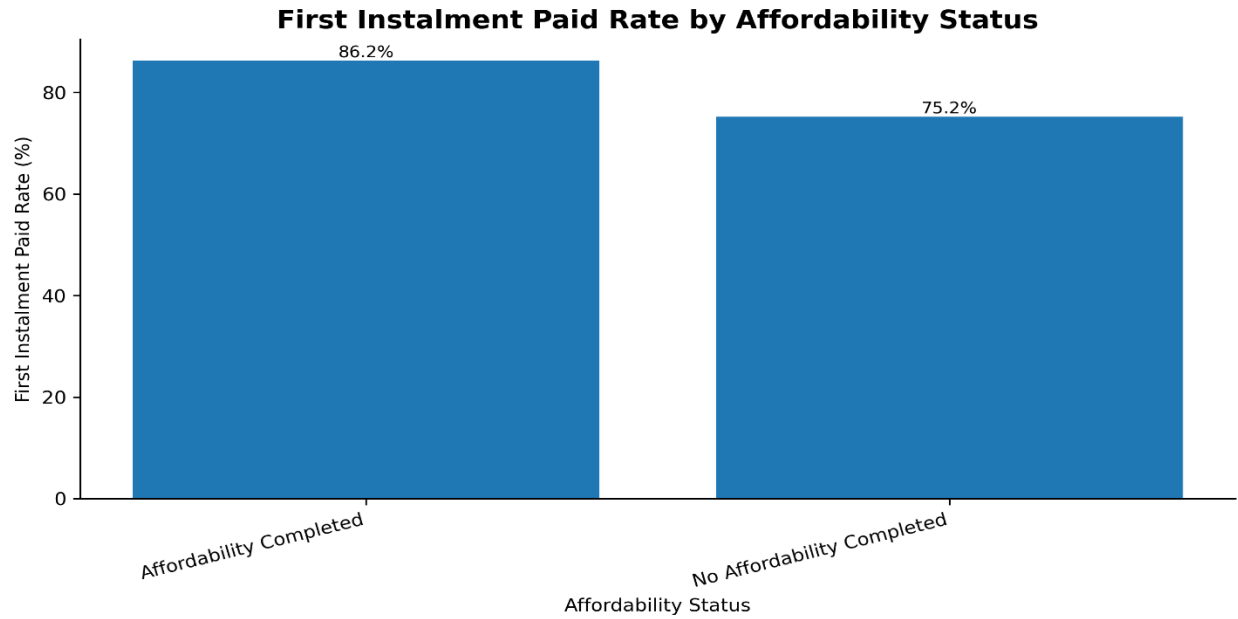
Recovery rate declines consistently as debt gets older: 0-30 days (6.10%), 31-90 (4.90%), 91-180 (3.08%), 181-365 (2.81%), and 365+ (1.48%). The sharpest decline occurs between the 0-30 and 31-90 bands, confirming that early intervention is critical for maximising recovery.



Business Takeaway: Customers in the 0-30 day debt age band recover at 4 times the rate of customers with debt older than 365 days. Prioritising newly uploaded accounts for immediate digital outreach and rapid journey entry would materially improve overall portfolio recovery.

Chart 10: Affordability Impact on Plan Quality

Customers who completed the affordability assessment paid their first instalment at 86.21% vs 75.19%, an 11pp advantage. Broken plan rates were 31.53% for affordability-completed customers vs 43.89%, a 12.36pp improvement in plan sustainability.



Business Takeaway: Affordability completion is the strongest predictor of plan sustainability in the dataset. Only 203 of 10,000 customers completed the affordability assessment (2.03%), meaning the vast majority of the available benefit remains unrealised. Scaling completion through UX improvements and proactive prompts represents a high-impact, low-cost intervention.

7. Phase 3: Power BI: Data Model and Interactive Report

7.1 Data Model

The dashboard is built from three BigQuery reporting views. Fact Customer and Fact AB Test are deliberately kept disconnected because the A/B test is a filtered experiment population; connecting them would corrupt totals. The three disconnected helper tables are controlled entirely by SWITCH measures and require no relationships.

Table	Source	Rows	Role
Fact Customer	vw_customer_summary (BigQuery)	10,000	Main dynamic table for Pages 1 and 3; drives all funnel, recovery and segment visuals
Fact AB Test	vw_ab_test_customer_outcomes (BigQuery)	1,936	A/B test experiment table for Page 2; disconnected from Fact Customer
Monthly Funnel Trend	vw_funnel_summary (BigQuery)	4,360	Aggregated monthly rates for Page 1 line chart
Funnel Stages	DAX DATATABLE	8	Disconnected helper table driving dynamic funnel bar chart and drop-off chart via SWITCH
AB Core Metrics	DAX DATATABLE	5	Disconnected helper table driving Control vs Variant clustered bar chart via SWITCH
AB Lift Metrics	DAX DATATABLE	5	Disconnected helper table driving Variant Lift waterfall chart via SWITCH
Key Measures	Calculated container	n/a	Holds all 60 DAX measures; Dummy column hidden

7.2 DAX Measures: 60 Measures Across 7 Groups

Measure Group	Count	Key Measures
Core Customer and Recovery	10	Total Customers, Total Outstanding Balance, Total Amount Collected, Overall Recovery Rate, Payment Conversion Rate, Plan Setup Rate, Broken Plan Rate
Funnel Stage Counts and Rates	11	Uploaded through Plan Created Customers (8 count measures), Invite Delivered Rate, Digital Journey Start Rate
Dynamic Funnel SWITCH	4	Funnel Customers, Previous Funnel Customers, Stage Conversion Rate, Stage Drop-off Rate

Measure Group	Count	Key Measures
A/B Test Measures	14	AB Test Customers, Control Customers, Variant Customers; AB rate measures and customer counts for all 5 outcome metrics
Control vs Variant	15	Control and Variant rates for all 4 metrics; 5 lift measures including Variant Amount Collected Lift
Dynamic A/B SWITCH	2	AB Selected Metric Rate (drives clustered bar), AB Selected Lift (drives waterfall)
Plan Sustainability and Trend	4	First Instalment Paid Customers and Rate; Monthly Link Open Rate, Monthly Payment Conversion Rate, Monthly Plan Setup Rate

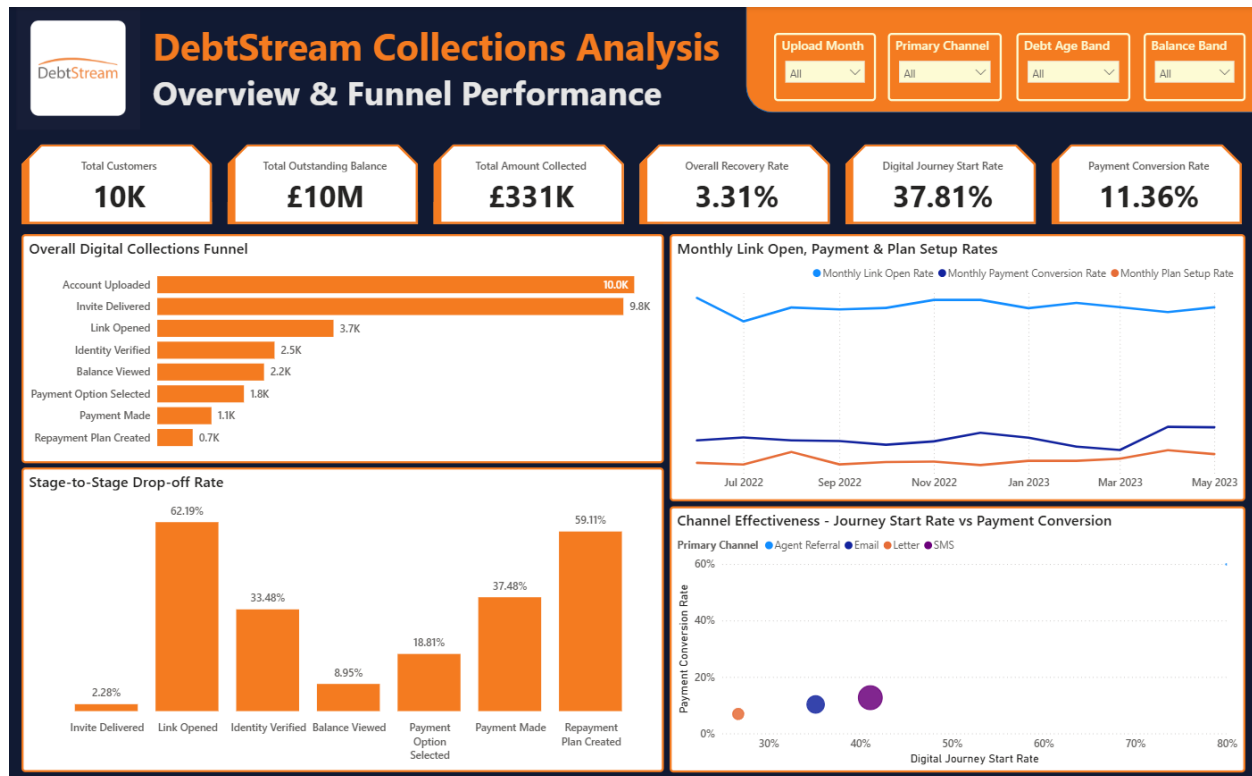
7.3 Report Pages

Page 1: Overview and Funnel Performance

Business question: How is the digital collections journey performing overall, and where are customers dropping off?

6 KPI Cards: Total Customers | Total Outstanding Balance | Total Amount Collected | Overall Recovery Rate | Invite Delivered Rate | Digital Journey Start Rate

4 Slicers (dropdown): Upload Month | Primary Channel | Balance Band | Debt Age Band (all from Fact Customer)



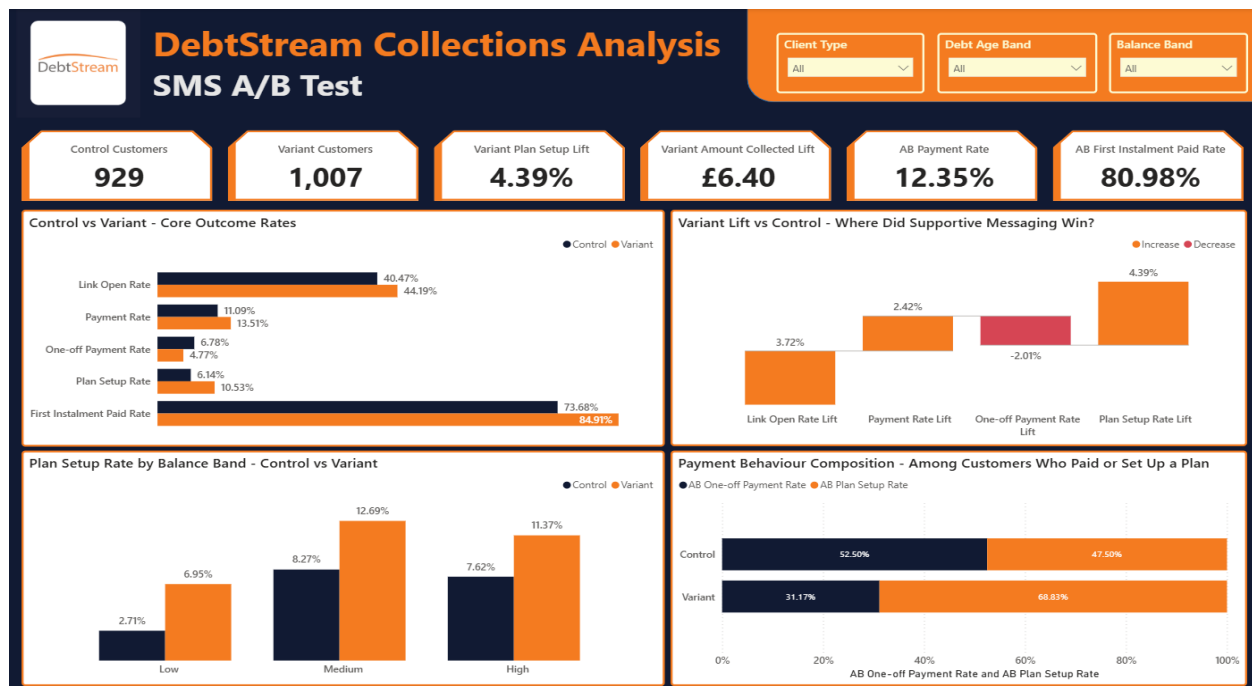
Visual	Type	Fields	Business Question
Overall Digital Collections Funnel	Horizontal Bar Chart	Funnel Stages[Stage] x [Funnel Customers], sorted by Stage Order	How many customers reached each journey stage?
Stage-to-Stage Drop-off Rate	Column Chart	Funnel Stages[Stage] x [Stage Drop-off Rate], Account Uploaded excluded	Which stage creates the biggest customer loss?
Monthly Engagement and Conversion	Line Chart	Monthly Funnel Trend[upload_month] x [Monthly Link Open Rate], [Monthly Payment Conversion Rate], [Monthly Plan Setup Rate]	Are outcomes improving or declining over time?
Channel Effectiveness	Scatter Chart	X: [Digital Journey Start Rate], Y: [Payment Conversion Rate], Size: [Total Customers], Detail: primary_channel	Which channel drives the best combined engagement and conversion?

Page 2: SMS A/B Test and Messaging Strategy

Business question: Did supportive SMS messaging outperform the direct payment reminder, and for which customer segments?

6 KPI Cards: Control Customers | Variant Customers | Variant Plan Setup Lift | Variant Amount Collected Lift | AB Payment Rate | AB First Instalment Paid Rate

3 Slicers (dropdown): Balance Band | Debt Age Band | Client Type (all from Fact AB Test)



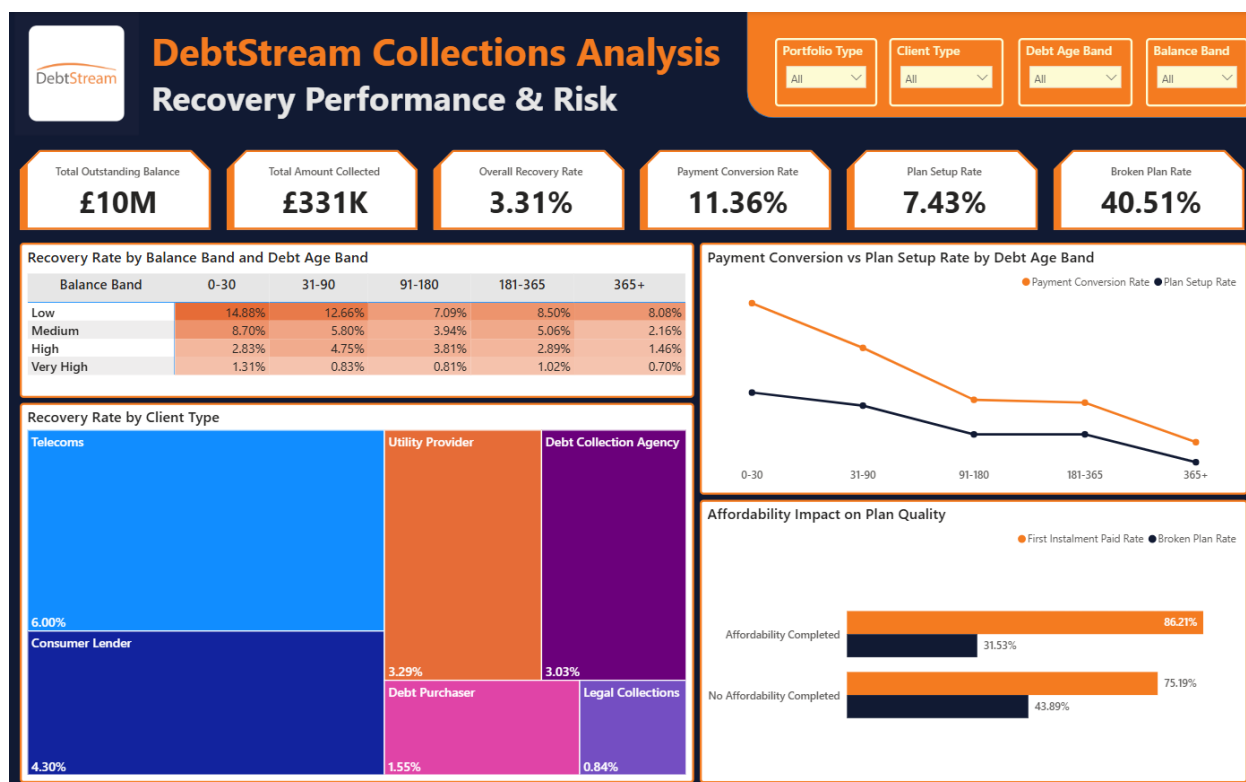
Visual	Type	Fields	Business Question
Control vs Variant Core Outcome Rates	Clustered Bar Chart (horizontal)	AB Core Metrics[Metric] x [AB Selected Metric Rate], Legend: ab_test_group, Control=grey, Variant=orange	Which message performed better across all key behaviours?
Variant Lift vs Control	Waterfall Chart	AB Lift Metrics[Metric] x [AB Selected Lift], negative lift visible for One-off Payment Rate	Where did supportive messaging win and where did it lose?
Plan Setup Rate by Balance Band	Clustered Column Chart	Fact AB Test[balance_band] x [AB Plan Setup Rate], Legend: ab_test_group	Which balance segments respond best to supportive messaging?
Payment Behaviour Composition	100% Stacked Bar Chart	Fact AB Test[ab_test_group] x [AB One-off Payment Rate] and [AB Plan Setup Rate]	How does each group split between one-off payments and plans?

Page 3: Segment Recovery and Risk

Business question: Which segments recover best, which are highest risk, and where should DebtStream prioritise action?

6 KPI Cards: Total Outstanding Balance | Total Amount Collected | Overall Recovery Rate | Payment Conversion Rate | Plan Setup Rate | Broken Plan Rate

4 Slicers (dropdown): Balance Band | Debt Age Band | Client Type | Portfolio Type (all from Fact Customer)



Recovery Rate by Balance Band and Debt Age Band

Balance Band	0-30	31-90	91-180	181-365	365+
Low	14.88%	12.66%	7.09%	8.50%	8.08%
Medium	8.70%	5.80%	3.94%	5.06%	2.16%
High	2.83%	4.75%	3.81%	2.89%	1.46%
Very High	1.31%	0.83%	0.81%	1.02%	0.70%

Recovery Rate by Client Type

Client Type	Recovery Rate
Telcoms	6.00%
Consumer Lender	4.30%
Utility Provider	3.29%
Debt Purchaser	1.55%
Debt Collection Agency	3.03%
Legal Collections	0.84%

Payment Conversion vs Plan Setup Rate by Debt Age Band

Debt Age Band	Payment Conversion Rate	Plan Setup Rate
0-30	11.36%	7.43%
31-90	~10.5%	~6.8%
91-180	~9.5%	~6.2%
181-365	~9.2%	~6.0%
365+	~8.5%	~5.5%

Affordability Impact on Plan Quality

Affordability Status	First Instalment Paid Rate	Broken Plan Rate
Affordability Completed	86.21%	31.53%
No Affordability Completed	75.19%	43.89%

Visual	Type	Fields	Business Question
Recovery Rate Heatmap	Matrix with Conditional Formatting	Rows: balance_band, Columns: debt_age_band, Values: [Overall Recovery Rate], red=low, green=high	Which balance and debt-age segment combinations are weakest?
Payment Conversion and Plan Setup by Debt Age	Line and Column Combo	debt_age_band x [Payment Conversion Rate] (column) and [Plan Setup Rate] (line)	Does recovery performance decline as debt gets older?
Recovery Rate by Client Type	Treemap	client_type x [Overall Recovery Rate]; block size shows relative recovery strength	Which client types recover most and least effectively?
Affordability Completion Impact on Plan Quality	Clustered Bar Chart (horizontal)	Fact Customer[Affordability Status] x [First Instalment Paid Rate] and [Broken Plan Rate]	Does affordability assessment improve plan sustainability?

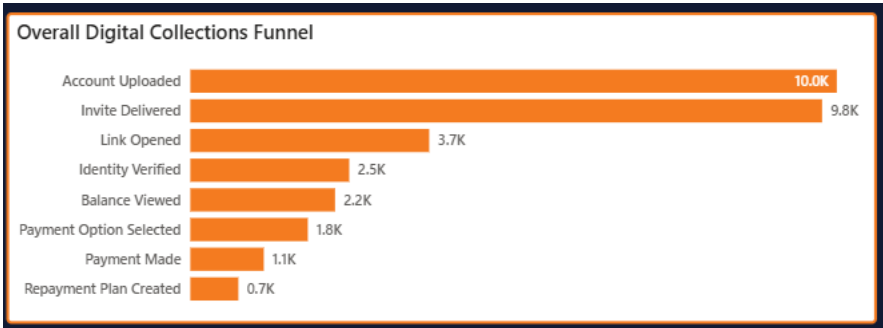
8. Key Findings and Business Insights

8.1 Funnel Performance

The Invite-to-Link-Opened Transition Is the Critical Failure Point

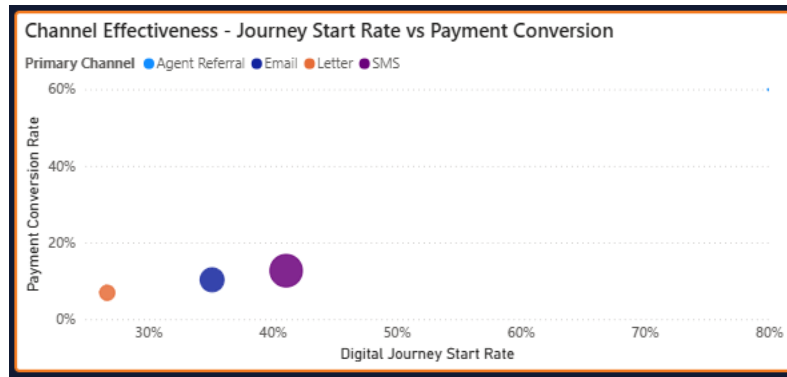
The customer journey begins with 10,000 uploaded accounts and 9,772 delivered invites, but only 3,695 customers open the link. The Invite Delivered to Link Opened transition loses 62.19% of customers, making it the single largest optimisation priority in the entire collections process. Once customers enter the journey, subsequent stages retain them at a far higher rate, confirming that the problem is journey entry rather than journey design.

Funnel Stage	Customers	Stage Drop-off Rate
Account Uploaded	10,000	n/a
Invite Delivered	9,772	2.28%
Link Opened	3,695	62.19%
Identity Verified	2,458	33.48%
Balance Viewed	2,238	8.95%
Payment Option Selected	1,817	18.81%
Payment Made	1,136	37.48%
Repayment Plan Created	743	59.11%



Channel Performance on Digital Engagement

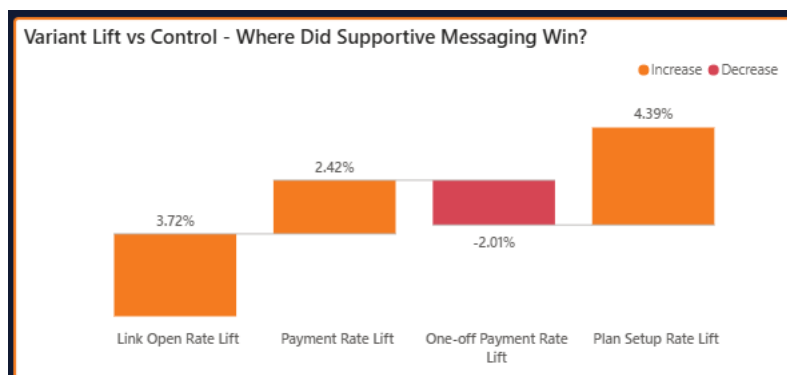
SMS-communicated customers start the digital journey at 41.11%, materially ahead of Email (35.14%) and Letter (26.70%). SMS also represents 59.22% of the total customer base, so its volume advantage amplifies its impact on overall funnel performance. The scatter chart on Page 1 confirms SMS leads simultaneously on both digital journey start rate and payment conversion rate.



8.2 SMS A/B Test Results

Variant Outperforms Control on All Plan-Based Metrics

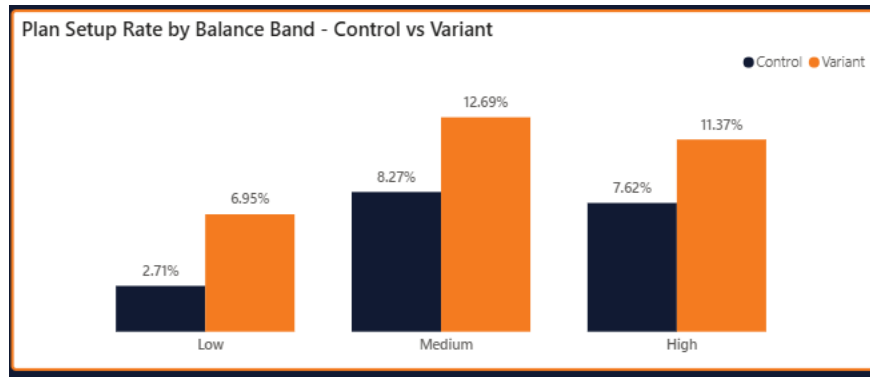
The supportive repayment SMS message (Variant) outperformed the direct payment reminder (Control) on four of five outcome metrics. Variant customers set up repayment plans at 10.53% vs 6.14% for Control, a statistically significant lift of 4.39 percentage points. Variant customers also collected £6.40 more per customer on average (£36.35 vs £29.95). The only metric where Control outperformed Variant was One-off Payment Rate (Control 6.78% vs Variant 4.77%), confirming that supportive messaging deliberately shifts customers toward structured plans.



Statistical significance: Two-proportion z-tests confirmed that plan setup lift and first instalment paid rate lift are statistically significant (p less than 0.05) overall, and particularly for Low and Medium balance segments. The A/B test results support a segmented messaging strategy rather than a single universal winner.

Variant Outperformance Is Consistent Across All Balance Segments

Variant outperforms Control on plan setup rate across all three tested balance bands: Low (+4.24pp), Medium (+4.42pp), High (+3.75pp). This consistency confirms the supportive messaging effect is not driven by a single segment but is a robust portfolio-wide finding. Medium balance customers show the strongest absolute response and should be prioritised for immediate rollout.



8.3 Recovery and Segment Performance

Balance Band: Low Balance Customers Recover Best

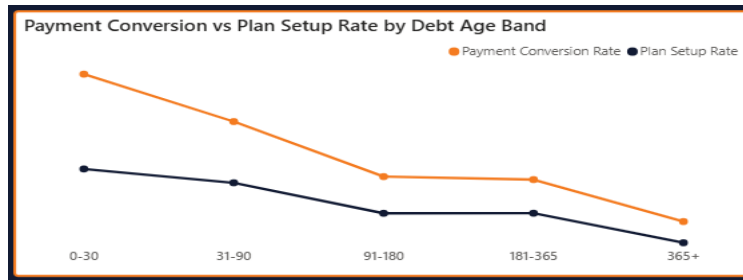
Recovery rate declines sharply as balance increases. Low balance customers recover at 10.57%, more than three times the overall portfolio rate of 3.31%. Very High balance customers recover at just 0.89% and have a broken-plan rate of 72.22%, representing the highest combined commercial risk in the portfolio.

Balance Band	Customers	Outstanding Balance	Recovery Rate	Broken Plan Rate
Low	3,508	£519,709	10.57%	41.32%
Medium	4,065	£2,549,618	5.00%	35.70%
High	1,853	£3,735,004	3.22%	48.99%
Very High	574	£3,192,271	0.89%	72.22%

Recovery Rate by Balance Band and Debt Age Band					
Balance Band	0-30	31-90	91-180	181-365	365+
Low	14.88%	12.66%	7.09%	8.50%	8.08%
Medium	8.70%	5.80%	3.94%	5.06%	2.16%
High	2.83%	4.75%	3.81%	2.89%	1.46%
Very High	1.31%	0.83%	0.81%	1.02%	0.70%

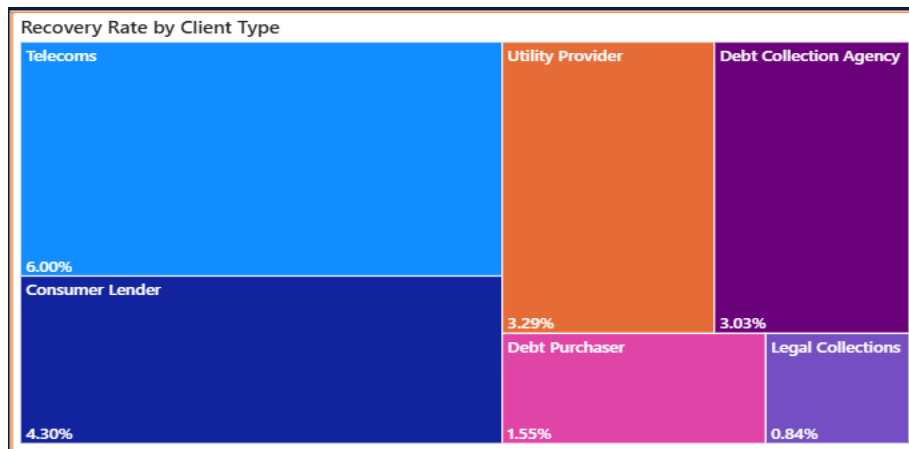
Debt Age: Early Intervention Is Critical

Recovery rate for 0-30 day accounts is 6.10%, four times higher than accounts aged 365+ days (1.48%). Payment conversion follows the same pattern: 18.00% for 0-30 day accounts declining to 5.97% for 365+ day accounts. The combo chart on Page 3 shows both rates declining in near-parallel as debt ages, confirming that the window of opportunity for digital collections narrows consistently with time.



Client Type: Telecoms and Consumer Lenders Lead Recovery

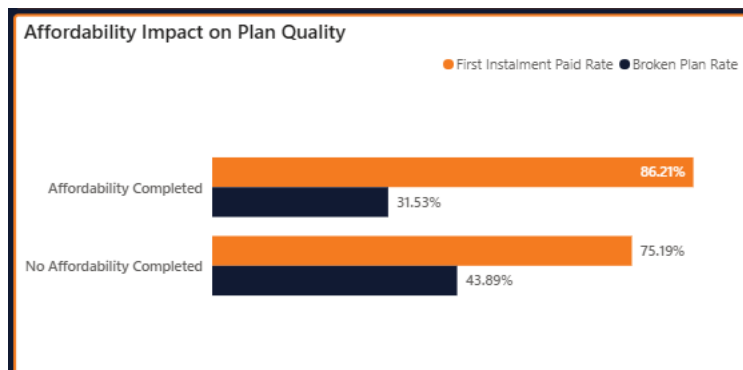
Telecoms clients show the strongest recovery rate at 6.00%, followed by Consumer Lenders at 4.30% and Utility Providers at 3.29%. Legal Collections show the weakest recovery at 0.84%, and Debt Purchasers at 1.55%. DebtStream should avoid using a single benchmark across all client portfolios because client type has a material impact on recovery outcomes.



8.4 Affordability and Plan Sustainability

Affordability Completion Is the Strongest Plan Quality Predictor

Customers who completed the affordability assessment paid their first instalment at 86.21% vs 75.19%, an 11.02 percentage point improvement. Their broken plan rate was 31.53% vs 43.89%, a 12.36 percentage point improvement in plan sustainability. Both effects are substantial. However, only 203 of 10,000 customers (2.03%) completed the affordability assessment, meaning the vast majority of the available benefit is currently unrealised.



9. Results and Recommendations

The analysis confirms five structural opportunities: a critical drop-off at the digital journey entry point, a validated superior SMS messaging strategy ready to scale, a strong relationship between debt age and recovery outcomes, a balance band segmentation imperative for collections strategy, and a significant untapped benefit from affordability completion. The following recommendations are presented in priority order based on expected commercial impact.

Recommendation 1: Fix the Digital Journey Start Problem First

62.19% of customers who received an invite never clicked through. This is the highest-priority intervention in the entire dataset. A/B testing invite message content, subject line copy, delivery timing, urgency framing and trust-building language should be the immediate focus. A 10 percentage point improvement in link-open rate from the Invite Delivered base of 9,772 would put approximately 977 additional customers into the digital journey. At current funnel conversion rates this generates approximately 111 additional payments and 73 additional repayment plans per upload cohort.

Recommendation 2: Adopt a Segmented SMS Messaging Strategy

The A/B test does not support replacing the direct reminder for all customers. It supports using supportive repayment-options messaging for customers who need flexibility (especially Medium and higher-balance segments) and maintaining direct payment reminders for customers with higher immediate payment readiness. Statistical significance testing confirms the plan setup lift is real and not directional noise. Variant should be adopted as the default SMS strategy and Control retained as an option for specific high-readiness customer profiles.

Recommendation 3: Treat Balance Band as the Core Segmentation Variable

Low-balance accounts are best suited to direct payment journeys with simple one-off payment options. Medium-balance accounts should receive stronger repayment plan messaging and nudges. High and Very High balance accounts need affordability-led journeys, stronger support prompts and closer plan monitoring because recovery is materially weaker and broken-plan risk is substantially higher. The 72.22% broken-plan rate for Very High balance accounts represents a specific and urgent problem requiring dedicated intervention.

Recommendation 4: Prioritise Early Intervention Before Debt Ages

0-30 day accounts recover at 6.10% vs 1.48% for 365+ day accounts, a 4x multiple. Operational prioritisation of newly uploaded accounts for immediate digital outreach and rapid journey entry would produce the highest return on collections effort. High and Very High balance accounts with debt over 180 days old likely require specialist intervention beyond the standard digital self-service flow.

Recommendation 5: Scale Affordability Assessment Completion

Only 2.03% of customers complete the affordability assessment, yet those who do show 11pp higher first instalment rates and 12pp lower broken plan rates. UX improvements to the affordability journey including reducing friction, clearer signposting, and proactive prompts within the digital flow should be prioritised. Even increasing completion from 2% to 5% would materially improve plan sustainability across the portfolio.

Recommendation 6: Track Plan Sustainability, Not Only Plan Creation

A repayment plan should not be treated as the final success metric. First instalment paid rate (78.20% across all plan customers), broken-plan rate (40.51%), missed payment counts and affordability completion should all be tracked alongside plan setup. This is especially critical for higher-balance accounts where plan sustainability is structurally weaker. DebtStream's product value depends not only on creating plans but on ensuring those plans are sustained to completion.

Results Summary

Finding	Current State	Recommended Action
Invite-to-Link-Opened drop-off	62.19%, largest single conversion failure in the funnel	A/B test invite content, timing, urgency framing and channel sequence
SMS messaging strategy	Control still deployed despite statistically significant Variant superiority	Adopt Variant as default; retain Control for high payment-readiness profiles
Balance band vs recovery	Low balance 10.57% recovery vs Very High at 0.89%	Segment strategy: Low=direct payment, Medium=plan nudge, High/Very High=affordability-first
Broken plan rate (Very High)	72.22% broken-plan rate on Very High balance segment	Specialist plan monitoring and stronger affordability support for Very High balance accounts
Debt age vs recovery	0-30 day recovery 4x higher than 365+ day	Prioritise newly uploaded accounts for immediate digital outreach
Affordability completion	2.03% completion rate, 97.97% of potential benefit unrealised	Reduce affordability journey friction and add proactive prompts within digital flow
Telecoms recovery (6.00%)	Best-performing client type, above portfolio average	Analyse and replicate Telecoms client operational approach across other portfolios
Legal Collections (0.84%)	Worst-performing client type, below portfolio average	Review digital suitability and consider specialist recovery routing for Legal Collections
Plan sustainability tracking	Plan setup tracked; first instalment and broken plan under-monitored	Track first instalment paid rate and broken plan rate as primary plan quality KPIs

10. Appendix: Screenshot Placeholders and GitHub Notes

The final GitHub version of this report should have screenshots inserted in the placeholders throughout before exporting to PDF. Each screenshot should directly support the business insight described in the surrounding text. The report should not become a chart dump; only the most impactful visuals from the Power BI dashboard and Python notebooks should be included.

Screenshot Placement Guide

Section	Placeholder	Recommended Screenshot
Section 6.1	Python Chart: Overall Digital Collections Funnel	Funnel bar chart from Notebook 1 showing 9 stages and absolute customer counts
Section 6.1	Python Chart: Stage-to-Stage Drop-off Rate	Drop-off column chart from Notebook 1 showing 62.19% at Invite-to-Link-Opened
Section 6.1	Python Chart: Digital Journey Start Rate by Primary Channel	Bar chart from Notebook 1 comparing SMS, Email, Letter and Agent Referral
Section 6.1	Python Chart: Monthly Funnel Trend	Line chart from Notebook 1 showing 12 months of link open, payment and plan rates
Section 6.2	Python Chart: Control vs Variant Core Outcome Rates	Side-by-side bar chart from Notebook 2 showing all 5 metrics for both groups
Section 6.2	Python Chart: Variant Lift vs Control by Metric	Lift bar chart from Notebook 2 with positive and negative bars clearly visible
Section 6.2	Python Chart: Plan Setup Rate by Balance Band	Clustered bar chart from Notebook 2 showing Variant vs Control by balance band
Section 6.3	Python Chart: Recovery Rate by Balance Band	Column chart from Notebook 3 showing recovery declining Low to Very High
Section 6.3	Python Chart: Recovery Rate by Debt Age Band	Column chart from Notebook 3 showing recovery declining 0-30 to 365+
Section 6.3	Python Chart: Affordability Completion Impact	Clustered bar from Notebook 3 comparing first instalment rate and broken plan rate
Section 7.3	Power BI Page 1	Full page screenshot of Overview and Funnel Performance dashboard
Section 7.3	Power BI Page 2	Full page screenshot of SMS A/B Test and Messaging Strategy dashboard
Section 7.3	Power BI Page 3	Full page screenshot of Segment Recovery and Risk dashboard
Section 8.1	Power BI: Overall Funnel Horizontal Bar	Funnel visual from Power BI Page 1 showing all 8 stages with customer counts
Section 8.1	Power BI: Channel Effectiveness Scatter	Scatter chart from Power BI Page 1 showing 4 channels plotted by two rates

Section	Placeholder	Recommended Screenshot
Section 8.2	Power BI: Variant Lift Waterfall	Waterfall chart from Power BI Page 2 showing lift with negative One-off Payment bar
Section 8.2	Power BI: Plan Setup by Balance Band	Clustered column from Power BI Page 2 showing Variant vs Control by balance band
Section 8.3	Power BI: Recovery Rate Heatmap	Matrix from Power BI Page 3 with conditional formatting showing red/green by segment
Section 8.3	Power BI: Payment Conv and Plan Setup Combo	Line and column combo from Power BI Page 3 showing both rates by debt age
Section 8.3	Power BI: Recovery by Client Type Treemap	Treemap from Power BI Page 3 showing recovery rate by client type block size
Section 8.4	Power BI: Affordability Impact Clustered Bar	Horizontal bar from Power BI Page 3 comparing affordability completion status

Final Project Conclusion

The DebtStream project demonstrates a complete analytics workflow from raw collections data through SQL modelling, Python analysis, statistical testing and Power BI reporting. The strongest business conclusion is that DebtStream should measure success not by delivered reminders or plan creation alone, but by customer journey entry rate, payment success rate, sustainable repayment plan rate, and recovery rate together. A segmented collections strategy driven by balance band, debt age, client type, messaging approach and affordability status is the most commercially actionable output of this analysis.